

Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru-3bb728174 | github.com/zac-09 | Kampala, Uganda

PROFESSIONAL SUMMARY

Senior Backend Engineer with 4+ years of concentrated, high-ownership experience building scalable Node.js systems on AWS and GCP. Led zero-downtime migration of 20+ production applications from Parse/MongoDB to Firebase, saving \$5,000/month in infrastructure costs. Proven expertise in API development, high-volume data processing (2M+ records), and microservices architecture with Apache Kafka and Kubernetes. Experienced in TypeScript/JavaScript, database optimization, and real-time systems for fintech and enterprise applications.

CORE COMPETENCIES

Node.js & TypeScript

AWS & Cloud Infrastructure

Zero-Downtime Migrations

High-Volume Data Pipelines

API Development & gRPC

Microservices Architecture

PostgreSQL & NoSQL Databases

Scalable System Design

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer — US Remote

- **Architected and executed zero-downtime migration** of 20+ production applications from Parse/MongoDB to Firebase/GCP, reporting directly to the CTO
- **Built real-time bidirectional sync pipeline** between MongoDB and Firestore using Node.js and Google Pub/Sub for high-volume message processing
- **Reduced infrastructure costs by \$5,000/month** by migrating all services from AWS to GCP
- Implemented 3D visualization feature using video overlay and ffmpeg, increasing buyer conversion rates
- Developed Express Shipping feature generating additional \$40 revenue per order

MTailor

Mar 2022 – Jul 2022

Backend Contractor — US Remote

- Built scalable data migration scripts to process and transform MongoDB collections to Firestore with complex business logic
- Created technical documentation for Firebase SDKs (Node.js/Python) for engineering team onboarding
- Trained operations team on Firebase Dashboard, backup procedures, and database management

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer — Singapore Remote

- **Built high-volume data pipelines processing 2M+ Firestore records** for real-time stock market price delivery from Morningstar APIs
- Developed responsive UI pages for PWA using React and Tailwind CSS
- Extended backend APIs with Firebase Cloud Functions and Express.js on Heroku

Mind2matter.co

Jan 2022 – Jul 2022

Software Engineer — US Remote

- Built backend services for DeFi applications using Web3 and Node.js
- Delivered React UIs from Figma designs under tight deadlines, reporting directly to CTO

CodeBits

Jan 2020 – Jul 2021

Software Engineer & Team Lead — Uganda Remote

- **Led team of 4 developers** building legal tech systems for NGOs in Uganda
- **Architected microservices backend** using Apache Kafka, Docker, and Kubernetes for FIDA Uganda's case management system
- Built cross-platform mobile app using React Native and Expo with Firebase push notifications
- Developed USSD service for legal aid providers with gRPC backend in Node.js

PROJECTS

Parse → Firebase Migration System Production

Real-time bidirectional sync pipeline for zero-downtime migration of 20+ applications. Built with Node.js, Google Pub/Sub, and custom data transformation logic.

Node.js, Firebase, GCP Pub/Sub, MongoDB, Python

Stock Price Data Pipeline Fintech

High-volume batch processing system handling 2M+ records for real-time stock market data delivery with Morningstar API integration.

Node.js, Firebase, Firestore, Express.js

Legal Aid Platform Microservices

Event-driven microservices architecture for case management system serving NGOs in Uganda. Includes USSD service for feature phone access.

Node.js, Apache Kafka, Docker, Kubernetes, gRPC, React Native

EDUCATION

Bachelor of Science in Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

SKILLS

Backend: Node.js, TypeScript, JavaScript, Python, Express.js, gRPC **Cloud:** AWS (S3, EC2, EBS), GCP, Firebase, Cloud Functions

Databases: PostgreSQL, MongoDB, Firestore, SQL **Infrastructure:** Docker, Kubernetes, Apache Kafka, Google Pub/Sub

Frontend: React, React Native, Tailwind CSS, HTML/CSS **Other:** Git, Linux, REST APIs, Microservices, Serverless